Assignment - Nanomaterials

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CSE A1

Q1 Describe in detail structural, optical, physical, chemical, electrical properties of nanomaterials

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Optical properties -

Because of the quantum confinement in nano-materials, the emission of visible light can be tuned by varying the nanoscale dimensions. It is observed that the size reduction in nanomaterials shifts the emission of peak towards the shorter wavelength (blue shift)

These properties are
(i) scattering
(ii) absorption

Bulk metals absorb electromagnetic radiation. Thin films of metal may partially transmit, just because there is insufficient material to absorb the radiation.
Apart from insufficient material effects, there are important phenomena which come into play in nanomaterials.
These include: dominance of surface plasmons, quantum confinement effects etc.

Structural properties-

Nanomaterials can exhibit unique properties

(i) Small size effect - The macroscopic size of the nanocrystalline unit constituting the nanosolid is in the nanometer range. The particle size is equal to or smaller than the light wavelength, the de Broglie wavelength

(ii) Macroscopic Quantum Tunneling effect - The ability of microscopic particles to penetrate the barrier is called the tunneling effect

(iii) Surface effect - refers to the phenomenon that the ratio of surface atom number to total atom number of nanoparticles increases sharply with the decrease in particle size.

Physical properties-

The melting point of a material directly correlates with the bond strength. In bulk materials, the surface to volume ratio is small and hence the surface effects can be neglected. However, in nanomaterials the melting temperature is size dependent and it decreases with the decrease particle size diameters.

Chemical properties-

(a) Increased to total surface area

(b) Increased number of atoms available on the surface

(c) Increased catalytic activity of these large number of surface atoms.

(d) Different / surface catalytic properties by the change in shape, size and composition

Electrical properties

- In bulk materials, conduction of electrons is delocalized, that is, electrons can move freely in all directions.
- When the scale is reduced to nanoscale, the quantum effect dominates. For zero dimensional nanomaterials, all the dimensions are at the nanoscale and hence the electrons are confined in 3-D space.

- For one dimensional nanomaterials, electrons confinement in 2-D space.

Q2 Explain various applications of Nanomaterials

A (i) ELECTRONICS

Carbon nanotubes are close to replacing silicon as a material for making smaller, faster and more efficient microchips and devices

(ii) ENERGY

Nanotechnology lowers costs, produces stronger and lighter wind turbines, improves fuel efficiency, hence can save energy

(iii) BIOMEDICINE

The properties of some nanomaterials make them ideal for improving early diagnosis and treatment of neurodegenerative diseases or cancer. They are able to attack cancer cells selectively without harming other healthy cells.

TEXTILE

Nanotechnology makes it possible to develop smart fabrics that don't stain nor wrinkle as well as stronger, lighter and more durable materials to make motorcycle helmets or sports equipment.